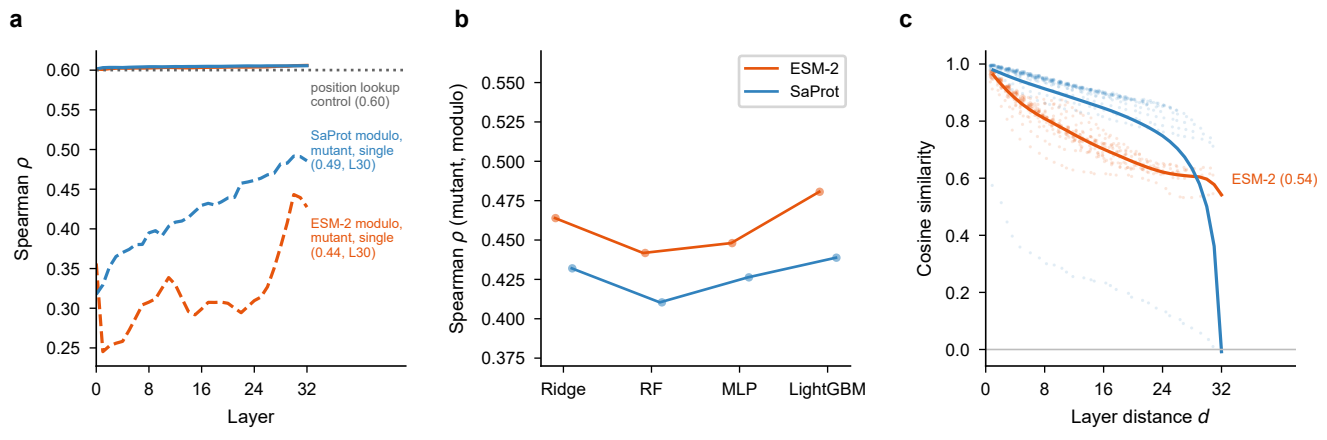

a
Spearman ρ
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
position lookup control (0.60)
SaProt modulo, mutant, single (0.49, L30)
ESM-2 modulo, mutant, single (0.44, L30)
0
8
16
24
32
Layer
b
Spearman ρ (mutant, modulo)
0.550
0.525
0.500
0.475
0.450
0.425
0.400
0.375
ESM-2
SaProt
Ridge
RF
MLP
LightGBM
c
Cosine similarity
1.0
0.8
0.6
0.4
0.2
0.0
ESM-2 (0.54)
0
8
16
24
32
Layer distance d